Capstone Project: Agentic Public Records Compliance Assistant

Selected Agent: Research Assistant

Checkpoint 4.1: Evaluation of Tree of Thought Reasoning

Is Tree of Thought Reasoning Appropriate?

For my Agentic Public Records Compliance Assistant, I have decided **not to implement Tree of Thought (ToT) reasoning in the current capstone design**.

The goal of my agent is to help public-records staff review documents, audio, and video requested under New Jersey's Open Public Records Act (OPRA). The agent identifies potentially sensitive information, retrieves relevant policies, recommends redactions with supporting evidence, and sends those recommendations to a human reviewer for approval.

In the earlier checkpoints, I introduced a ReAct-based reasoning design, short-term and long-term memory, and external tools. In Checkpoint 3.1, I demonstrated semantic retrieval using official NJ OPRA policy documents to support a recommendation involving a synthetic personal phone number.

Tree of Thought is useful when an agent needs to explore several possible reasoning paths before selecting a solution. Instead of following one sequence of decisions, it generates alternative possibilities, evaluates them, and explores the most promising paths.

I considered whether this approach would improve my agent's ability to evaluate complex redaction scenarios. However, I believe that **ToT would introduce more complexity than benefit for the current scope of my project**.

The primary challenge is not generating different interpretations of a situation. It is accurately detecting sensitive information, identifying applicable policies, verifying the supporting evidence, and avoiding unsupported legal conclusions.

A system that explores multiple possible interpretations without sufficient policy evidence could generate plausible but incorrect recommendations. For a compliance-related application, producing more possible answers does not necessarily mean producing a more reliable answer.

I therefore believe a focused reasoning workflow, supported by external retrieval and human review, is more appropriate for the current capstone.

Why ToT Is Not the Best Fit

I evaluated ToT against the types of problems my agent is expected to solve.

First, the initial redaction scenarios have a relatively narrow decision space.

For example, when a synthetic transcript contains a personal phone number, the agent needs to identify the information, retrieve the relevant policy, assess whether it supports redaction, and prepare a recommendation.

The agent does not need to generate and explore several competing solutions. It needs to establish whether reliable evidence supports an action.

The main outcomes are to recommend redaction, determine that available evidence does not support redaction, or refer the matter for human review.

Second, retrieval provides a stronger foundation than exploring alternative reasoning paths.

My agent relies on official NJ OPRA statutes and Government Records Council guidance. These documents provide the external evidence needed to support recommendations.

In Checkpoint 3.1, I demonstrated how the workflow retrieves policy information and uses it to support a proposed redaction of a synthetic personal phone number.

I also tested a missing-evidence scenario. When the retrieval tool could not establish an applicable rule, the workflow attempted another search and then escalated the item for human review without inventing a legal justification.

This behavior is particularly important because the agent should recognize when it does not have enough information to make a supported recommendation.

ToT would not solve missing or inadequate policy evidence. Exploring additional branches could increase the number of unsupported interpretations without improving the underlying evidence.

Third, ToT would increase implementation complexity and processing cost.

A ToT design would require generating multiple candidate reasoning paths, evaluating those candidates, maintaining their separate states, and deciding which path to pursue.

For each sensitive-information finding, this could result in several additional model calls and policy searches.

Since the capstone has a limited development timeline, I would prefer to invest that effort in retrieval quality, tool reliability, testing, and human feedback.

These capabilities are directly connected to the problem the agent is solving and can be evaluated through observable outcomes.

Finally, complex legal interpretations should not be resolved through AI-generated consensus alone.

Some public-records decisions may involve multiple laws, exemptions, or contextual considerations. In those cases, several possible interpretations may exist.

Although ToT could help organize those alternatives, it should not be used to select a legally authoritative conclusion simply because one reasoning branch receives a higher score.

For the current project, I would rather have the agent identify uncertainty, present relevant evidence, and request human review.

Preferred Reasoning Approach: ReAct with Retrieval and Human Feedback

The reasoning approach that best fits my project is **ReAct (Reasoning and Acting), supported by Retrieval-Augmented Generation (RAG), structured workflow state, and human-in-the-loop review**.

ReAct enables the agent to decide what action is required, execute an available tool, observe the result, and determine the next step.

For example, consider a synthetic record containing the statement:

"My personal phone number is 609-555-0142."

The workflow would proceed as follows:

1. **Understand the record:** The agent identifies the record type and the information that needs examination.
2. **Detect sensitive information:** It invokes a detection tool that identifies the personal phone number.
3. **Determine the next action:** The agent recognizes that detecting a phone number does not independently establish the legal basis for redaction.
4. **Retrieve policy evidence:** It searches the indexed NJ OPRA/GRC documents for relevant guidance.
5. **Evaluate the observation:** The agent examines the retrieved passage, verifies the source, and considers whether any exception or uncertainty requires review.

6. **Prepare a recommendation:** It produces a proposed redaction with supporting policy references.
7. **Obtain human feedback:** A reviewer approves, rejects, or modifies the recommendation.
8. **Continue or terminate:** The workflow updates the recommendation when needed and completes only after the required review.

This approach supports reliable behavior because each decision is connected to an observable tool result or retrieved source.

Memory and State Management

Short-term memory will retain the current request, detected findings, retrieved passages, tool outputs, recommendations, and reviewer feedback.

Long-term memory will provide access to indexed public policy documents, including their titles, dates, sections, and source references.

Maintaining this information helps the agent avoid repeating completed work and supports consistent decisions across multiple steps.

Handling Failures and Uncertainty

If a retrieval result is missing or insufficient, the agent can reformulate the query and retry within a defined limit.

If sufficient evidence is still unavailable, the workflow will stop making policy-based recommendations and escalate the finding for human review.

Similarly, if a tool fails, the workflow can retry or request assistance rather than continuing with incomplete information.

This produces a clearer and more auditable process than exploring several speculative reasoning branches.

Future Consideration for ToT

I would reconsider ToT if the project later expands into complex cases involving several overlapping exemptions, competing legal interpretations, or multiple possible disclosure strategies.

In that situation, a limited ToT process could help organize candidate interpretations for human review. Each candidate would still need authoritative supporting evidence.

However, this is outside the initial capstone scope.

Conclusion

I considered Tree of Thought reasoning and its potential benefits, but decided that it is not necessary for the current Public Records Compliance Assistant.

ReAct combined with semantic retrieval, structured memory, tool validation, and human review provides a simpler and more reliable approach for the problem I am solving.

My priority is to build an agent that can retrieve evidence, make explainable recommendations, recognize uncertainty, and safely involve a human when needed.

This approach also allows me to focus on testing observable behavior and improving the reliability of the capstone rather than adding reasoning complexity without a clear benefit.